

Reverb hiding: A new framework for audio steganography

Ali Erdem Altınbaş^a, Mehmet Zeki Konyar^{b,*}

^a Department of Electronics and Communications Engineering, Kocaeli University, İzmit, Turkey

^b Department of Software Engineering, Kocaeli University, İzmit, Turkey

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ABSTRACT

In the last decades, a lot of new approaches have been developed for audio steganography. However, most of these studies ignore musical hearing. In this paper, we proposed a new framework for hiding data in audio using the reverb effect, which is one of the time-dependent effects in music production. Reverb is an audio effect in music production that creates an artificial atmosphere that simulates what could be found in a natural setting such as a large hall, stadium, etc. The impulse response of the related environment is convolved with an audio file in a studio environment. This study proposes embedding secret data in audio signals with the reverb effect. The secret message is embedded into the impulse response of the natural environment and then the audio signal is reverbed with this impulse response. The original and reverberated audios are again blended for parallel processing, resulting in stego-audio. In the data extraction stage, firstly the fully reverberated audio is obtained from stego-audio which was blended with the original (dry) audio. Then, the reverberated audio file is deconvolved with the dry audio again, and the stego impulse response is obtained.

Existing studies of audio steganography such as echo hiding are common techniques used in music production but have a significant impact on hearing. The proposed reverb hiding is done with much more complex parameters, so the stego-audio is more compatible with real-life applications. The reverb-hiding achieves superior PSNR results for different sizes of the secret message. The proposed method evaluated in terms of oscillograms, frequency spectrum analysis, spectrograms, cepstrogram analysis, histogram analysis, and autocorrelation analysis. The experimental results are also compared with six different existing audio-tech methods. The proposed reverb-hiding method significantly impacts the trade-off between capacity and imperceptibility, as indicated by the results.

1. Introduction

In contemporary applied science, the employment of data obfuscation and digital imprinting has become more widespread in diverse fields. These practices include protecting private communications, advancing forensic techniques, managing metadata, and addressing copyright infringements [1]. In a manner similar to the interplay between encryption and decryption, steganography and steganalysis methods continue to evolve in a mutually beneficial way. Developing new steganography methods and steganalysis techniques is not only a theoretical endeavor, but also has practical uses [2]. Despite extensive research on audio security and data transmission, the risk of intruders discovering new methods to access sensitive information remains a constant threat. This creates a significant vulnerability for data compromise through targeted attacks. Current measures designed to protect Internet-based data exchange and user privacy are insufficient,

allowing for unauthorized interception. As a result, there is an urgent need to strengthen the security and confidentiality of data transfers. To address this issue, techniques such as steganography and cryptography can be utilized. This paper presents an innovative approach that combines reverb hiding with audio steganography, offering a robust and secure method for embedding hidden information in audio signals.

Steganography involves a carrier cover object and a secret message component, where the message is embedded into the carrier cover for a specific purpose. Typically, the carrier object can take the form of an image, audio, video, or document [3–5]. Most of the literature studies use the image as the carrier cover to be embedding the secret message [6]. However, there have been less studies conducted on the use of the audio as the carrier cover, which is often more susceptible to distortion [7]. Since this carrier is called a cover in data hiding terminology, the name of the file that will carry the message will be referred to in the rest of the paper as the cover audio. In this article, the term cover does not

* Corresponding author.

E-mail address: mzeki.konyar@kocaeli.edu.tr (M.Z. Konyar).

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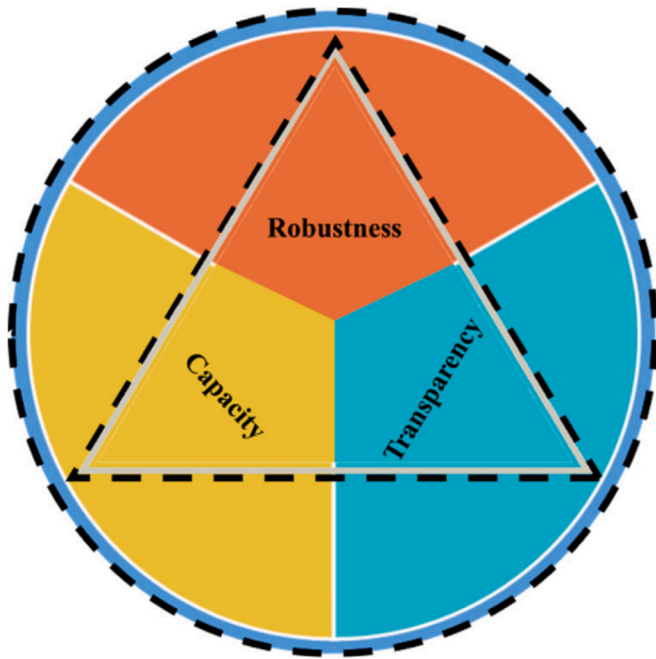


Fig. 1. Trade-off in steganography [12].

mean the same as the term used for a previously released song by someone other than the original artist.

audio steganography is one of the most important topics for data hiding where the generated audio file size increases remarkably with the developing technology. In data hiding, it is very important to keep the corruption of the carrier object to a minimum [8]. While visual distortion occurs in images and videos, auditory distortion occurs in sounds. The most critical aspect of hiding data in audio is that even low distortions can be noticed due to the high sensitivity of the human ear. For this reason, data hiding methods developed for voice are done with limited methods [9]. These methods consist of approaches made in the spatial domain and frequency domain. While LSB (least significant bit) coding, parity coding and echo hiding methods are generally used in the spatial domain, discrete wavelet transform, tone insertion and amplitude coding methods are used in the frequency domain. In addition to known methods, in recent years, artificial intelligence methods have become popular for data hiding. For example, Huu et al. developed a novel audio steganography technique using a convolutional neural network (CNN). The proposed deep neural network comprises of two sub-models: the preparation network extracts audio characteristics and concatenates them with the image, while the hiding network conceals secret audio features within the cover image to generate a container image. This approach enables hiding and retrieval of original audio data from encrypted images [10].

In this study, an innovative approach has been made to audio data hiding by hiding data within reverb signals. Convolution reverb is an audio effect that simulates the reverberation of a physical or virtual space. While most studies pay attention to data hiding methods, they are not sensitive enough to preserve the natural characteristics of the sound and act accordingly, while reverb signals are compatible with the natural structure of the sound.

The ability to perceive sound distortion as a property of sound is crucial in addressing this issue, particularly as studio recordings are typically made in isolated environments [11]. Subsequently, mix engineers employ various tools to achieve the intended sound for the dry signal. Based on the idea that certain tools used by mix engineers are well-suited for data hiding, this study introduces a framework for concealing audio using reverb.

The reverb is applied to a dry signal and expands the sound's

perceived spatial dimensions, making it seem as if it had been recorded in an alternate, distant environment. The most straightforward approach to achieve this effect is using the convolution reverb, which involves recording impulse responses (IR) from real-world locations using specific microphone techniques, and then convolving these responses with dry signals. The sound emanates from the environment where the IR of real-world was recorded, creating a natural effect in music production. Therefore, embedding data within the IR is crucial for the success of the proposed work.

In this paper, a novel data hiding method is proposed for data hiding in audio files by using the reverb effect. Reverb is an audio effect in music production that creates an artificial atmosphere that mimics what could be found in a natural setting. In the proposed method, this IR is convolved with an audio file that was recorded in a studio setting; this is referred to as convolution reverb in the music production industry. Stego-audio is produced by blending the original and reverberated sounds for parallel processing. Because the stego-audio was mixed with the dry audio, the fully reverberated audio is retrieved during the first stage of data extraction. The audio file is then separated from the dry audio. The proposed method focuses on minimizing the losses or changes that occur after embedding a secret message into the audio. PSNR, a key measure of quality degradation, is commonly used in literature to assess the impact of data hiding. In this study, the goal is to achieve the notably high PSNR values. This indicates that the reverb hiding technique achieves superior audio quality in terms of PSNR when compared to other methods. Furthermore, the proposed method aims to outperform state-of-the-art audio steganography techniques such as LSB, Echo, Wavelet, Spread Spectrum, and Cepstrum. To ensure the reliability of the results, the method also incorporates important audio reverb metrics used by sound engineers such as Reverb time (RT), Direct-to-reverberant ratio (DRR), correlation of energies in the first 50 and 80 ms (C50 and C80), early decay time (EDT), and just noticeable difference (JND), showing robust performance across these metrics. Various audio analysis graphs further demonstrate that the stego-audio remains largely unaffected by the proposed method, with only minor, acceptable changes observed in some cases. Overall, the proposed method ensures no significant changes in the main graphical results, reinforcing the effectiveness of the proposed approach in maintaining audio quality.

The following are the advantages of the proposed method and its contributions to literature:

- Unlike the classical methods used in literature, in this study the use of reverb is proposed to hide secret data in audio files.
- To best of our knowledge, this is the first study to hide data within the audio file using the reverb method, which is an artificial sound effect.
- Since the current technique inserts secret data into the reverb signal instead of sound, the distortion in the audio signal is reduced. So, the stego audios are compatible with real-life applications instead of the previous methods.
- When the results of the proposed study were compared with six different state-of-the-art audio steganography algorithms in the literature, superior performance was achieved compared to other audio data hiding methods.
- The performance of the proposed technique evaluated with different performance evaluation metrics which are used by sound engineers. The superior of the proposed method has also been proven with these metrics.

The rest of the paper is organized as follows: Section 2 explained the overview of the existing audio steganography methods and related works. The phase of the proposed method is explained in Section 3. Section 4 evaluates the proposed method's embedding capacity, imperceptibility, and quality. This section also includes comparisons with the methods from the literature. Section 5 provides the conclusion.



Fig. 2. User Interface of Galaxy Tape Echo by Universal Audio.

2. Related works

Data hiding involves three main phenomena: capacity, robustness, and transparency while embedding secret messages into the carrier files. As shown in Fig. 1, these phenomena have a trade-off with each other. Although there are various names used in the literature, the main objective of a steganography method is to hide as much data as possible in a way that is as undetectable as possible and robust against data manipulation methods, especially noise and compression [12]. While the robustness and the capacity are universal concepts across all types of covers, imperceptibility specifically refers to inaudibility in audio steganography [13].

Many studies that emphasize at least one of these three basic principles date back to before the 2000 s [14]. In the last 10 years, modifications of state-of-the-art methods, new methods using artificial neural

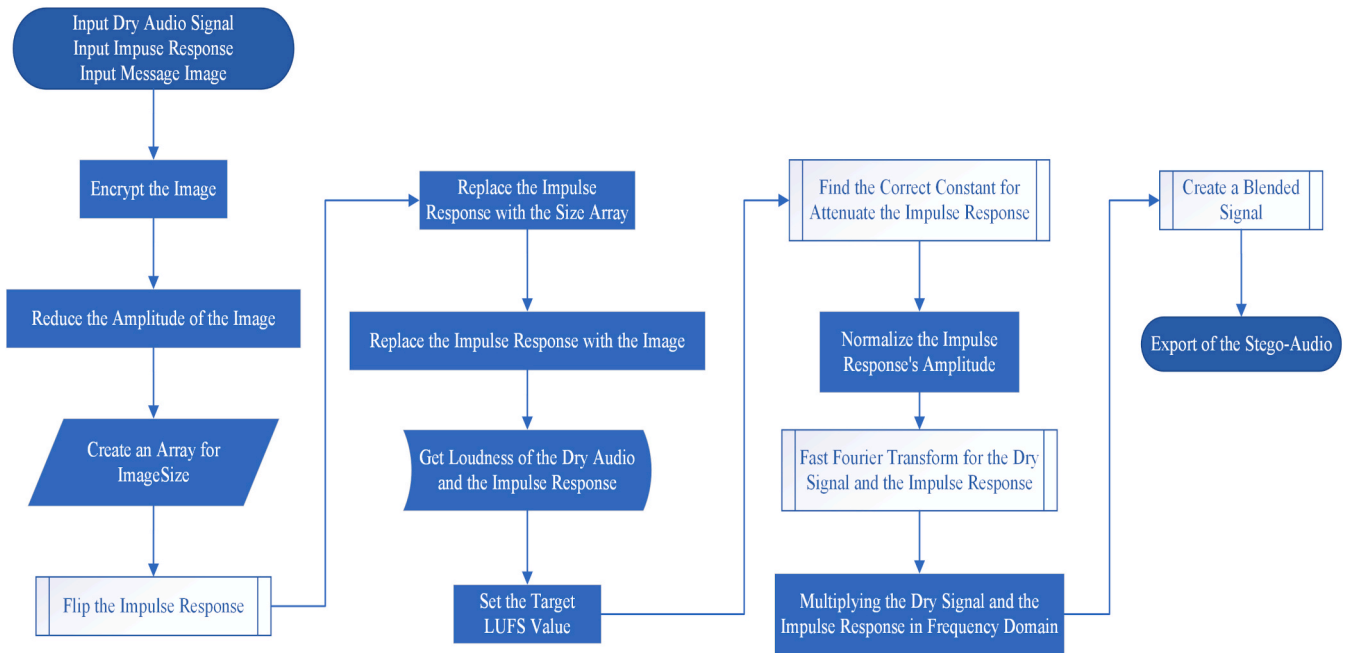


Fig. 3. Flowchart of proposed data hiding algorithm.

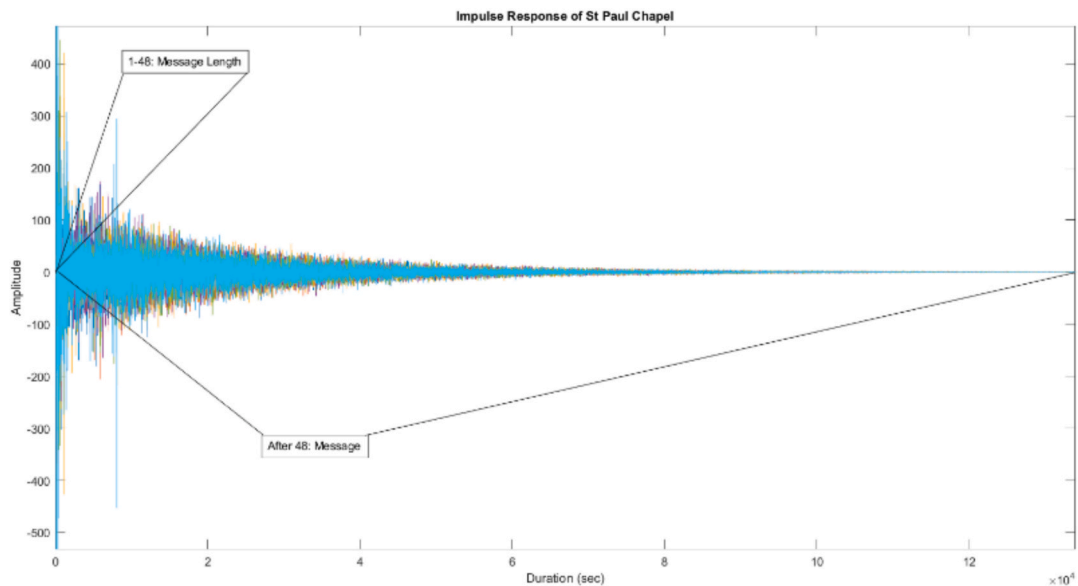


Fig. 4. Data hiding on IR.

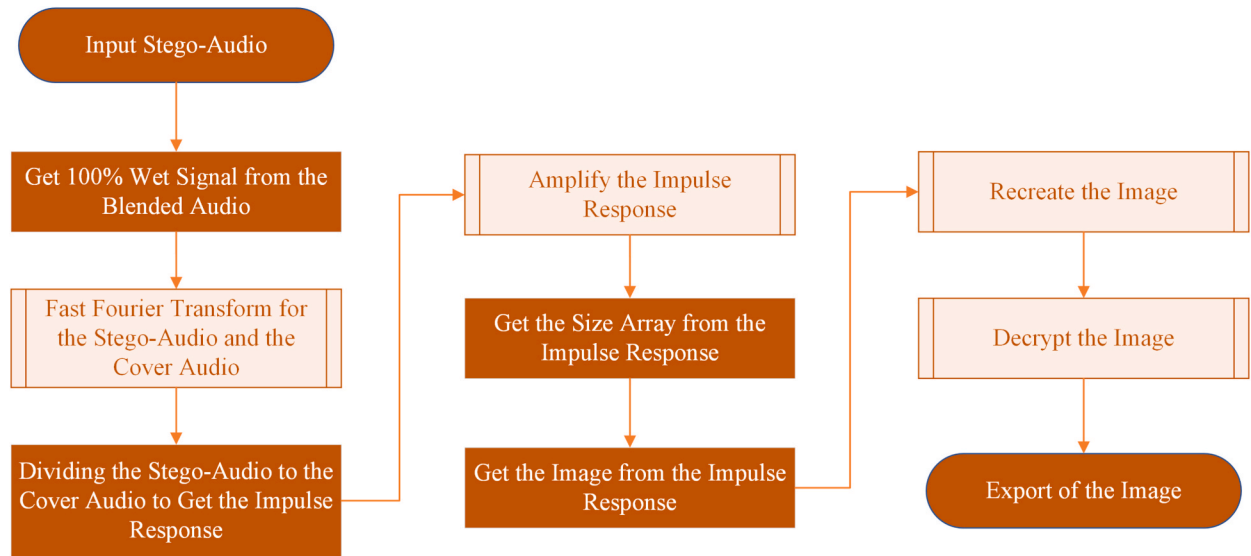


Fig. 5. Data extracting flowchart of the proposed algorithm.

networks, coverless steganography, etc. have been studied more frequently [15,16].

2.1. Least significant bit (LSB)

LSB technique, one of the most common methods in steganography, is frequently used in audio steganography [12]. Apart from the improvement of LSB algorithms, new studies also include topics such as the effect of cover audio [1]. The LSB method is widely used due to its easy applicability, but imperceptibility is a crucial principle to consider as it can be easily detected. Therefore, there is a common focus in the literature on developing less detectable methods with higher capacity on the LSB technique [17].

2.2. Spread spectrum

Spread spectrum technology is a data hiding technique that uses spreading sequences, also known as chip codes, to modulate the original signal in such a way that its energy is spread over a wide frequency spectrum. Chip codes can be generated using a variety of different methods, including nonlinear rules with a normal distribution, uniform random number generation over specified intervals, and pseudo-random bit generators. Moreover, Walsh-Hadamard chip codes, derived from Hadamard matrices, are commonly utilized as discrete sequences. These chip codes effectively modulate the original signal and serve as the key to embedding information within the scrambled data. To extract the hidden information, the receiver uses identical spreading sequences to demodulate the received signal and retrieve the hidden information through correlation analysis [18]. Researchers optimize the data hiding process by carefully designing and manipulating spreading sequences to ensure reliable and secure embedding and extraction of information within multimedia files such as audio [19].

2.3. Echo hiding

Echo hiding is another interesting method. It is specifically examined in this study because, like reverb, it is one of the time-based effects that the mix engineers commonly use. However, the most important issue in adding these effects is naturalness. In scenarios where artificial reflections such as reverb, echo, and delay are added to the original sound, these effects must be added carefully to maintain the naturalness [20].

Echo hiding essentially splits the cover audio into small pieces and adds echoes to each sample of those pieces [21]. The result of each piece

is defined as (1):

$$y[n] = x[n] + \alpha x[n - \gamma] \quad (1)$$

where $x[n]$ is the value of cover audio block, α is the echo decay rate and γ is the delay in seconds [22]. However, this approach was less effective than steganalysis methods and required improvement [23]. While many methods aim to achieve mathematical precision, it is also possible to perceive unnatural effects when listening. Because artificially creating an ambient effect is too complex to create with two parameters [24].

For example, Universal audio's Galaxy Tape Echo plug-in, shown in Fig. 2, emulates an analog tape echo device from the 1970 s. In this emulation, the condition of the cartridge can be selected as new, used, or old. The sound of the point where the two ends of the tape meet can be changed in this way. Also, the left and right balance of the echo sound and a simple equalizer can be adjusted independently. Many of such parameters contribute to the realistic sound of the echo effect.

Therefore, in contrast to conventional audio steganography methods, this paper presents a reverb hiding method inspired by a commonly used effect in music production.

3. The proposed reverb hiding method

3.1. Mathematical background of convolution reverb

Although artificial reverb dates to the 1960 s, convolution reverb, which became popular in the late 1990 s [25], is the algorithm used in our study. Convolution reverb is an audio effect that simulates the reverberation of a physical or virtual space with an IR of real-world. Depending on where the IR is received, this real-world environment can be concert halls, churches, rooms, or cathedrals. This effect is achieved by convolving the input audio signal with the IR of the simulated space. IR are recorded acoustic space responses when an ideal impulse or appropriate approximation is played.

During the song creation, it is essential to produce an artificial stage experience while maintaining its integrity. Therefore, parallel processing is necessary to apply various types of reverbs to different sounds in the desired amount. The initial sound recorded in an isolated studio environment is called the dry audio signal, while the reverberated version is known as the wet audio signal. The wet audio simulates an environment that is too dense to be heard clearly, with decay times of up to tens of seconds. The mix setting in reverb software allows for the blending of dry and wet sounds at the appropriate point to achieve the

Table 1
PSNR (dB) values of 32-bit depth floating-point Stego Audio for various TRs with different sizes of secret message.

Secret Message Size	PSNR (dB)				Birmingham Symphony Hall	Hermitage Theatre	Glazunov Concert Hall	Knights Hall Refectorium	London Arena	Luzit Cave	Lexicon 480L AutoPark	Lexicon 480L Large	San Francisco Columbarium Hall	Plate 250	JamSync Florida Room	The Wooden Recital Room
	Big Cave	Big Synagog	Big	Big												
1 kB	146.16	142.54	134.37	146.96	202.18	138.02	127.31	134.24	193.21	195.94	136.57	216.79	153.21	178.12		
5 kB	132.81	134.40	120.17	131.66	142.94	122.84	107.90	118.07	107.53	110.78	62.46	134.79	134.22	127.76		
10 kB	129.04	129.48	111.91	126.38	134.26	116.82	102.16	112.58	103.61	102.32	62.46	129.30	113.52	122.81		
15 kB	125.76	125.49	62.74	125.94	130.44	113.80	97.38	109.03	100.22	62.71	56.30	125.85	56.13	64.86		
20 kB	123.67	122.41	62.74	62.84	125.06	109.95	56.54	105.28	70.01	62.71	56.30	68.68	56.13	64.86		
25 kB	117.82	119.61	56.59	62.84	119.58	59.32	56.53	101.90	70.01	62.71	56.30	68.68	56.07	64.86		
30 kB	115.61	115.44	56.57	62.84	59.97	59.32	56.49	60.05	63.93	62.70	52.63	62.50	47.67	58.74		
35 kB	110.76	66.80	52.89	62.84	59.97	59.32	50.41	60.05	63.93	62.68	52.63	62.50	52.89	58.74		
40 kB	109.55	60.72	52.75	56.74	53.84	59.32	46.78	53.91	60.36	56.63	52.56	58.78	52.75	58.74		
45 kB	106.97	60.72	49.65	56.74	53.84	59.32	46.73	53.91	60.36	56.65	49.80	55.89	49.65	55.11		
50 kB	104.80	57.14	46.55	56.74	53.84	59.31	44.06	53.91	60.33	56.64	47.32	51.69	46.55	55.06		
55 kB	53.07	57.15	39.68	56.74	50.20	59.31	41.58	53.89	60.24	56.62	42.82	—	39.68	52.26		
60 kB	53.07	54.59	—	53.14	50.15	53.07	38.96	53.86	57.69	53.08	35.46	—	—	49.75		
65 kB	53.07	52.59	—	53.14	47.52	53.07	35.47	50.25	57.63	53.04	—	—	—	47.67		
70 kB	53.07	52.55	—	50.56	45.40	53.07	30.98	50.21	55.60	53.03	—	—	—	45.14		
75 kB	53.07	50.86	—	48.52	45.23	53.07	28.12	47.60	55.50	50.46	—	—	—	40.94		
80 kB	46.88	49.29	—	48.30	42.88	49.31	—	45.48	53.59	50.36	—	—	—	35.96		
85 kB	44.25	47.82	—	46.48	40.53	49.31	—	45.34	52.16	48.36	—	—	—	32.73		
90 kB	44.25	45.98	—	46.02	37.21	46.49	—	44.93	50.63	46.66	—	—	—	—		
95 kB	42.18	44.08	—	44.15	32.32	43.54	—	41.43	49.83	44.01	—	—	—	—		
100 kB	42.18	42.65	—	42.34	30.25	38.85	—	40.41	47.98	43.82	—	—	—	—		

desired ambient effect.

The audio effect is achieved by multiplying each sample of the input signal with the corresponding sample of the IR [26].

The reverb effect is created by convolving the dry audio signal (S_d) with the IR to obtain the reverberated signal (S_r) with the formula given in the following equation [27].

$$S_r = S_d \otimes IR \quad (2)$$

When this process is explicitly stated with more detailed equation as given in (3). It is possible to represent the convolution operation in the time domain with this equation.

$$y[i] = \sum_{j=0}^{M-1} h[j]x[i-j] \quad (3)$$

where y is the output, x is the input, i is the sample in the output system, j is the IR sample from 0 to $M - 1$, h is the IR, and [28]. However, performing all of these operations in frequency domain is much more computationally efficient [29] as follows:

$$Re(Y[f]) = Re(X[f]) \times Re(H[f]) - Im(X[f]) \times Im(H[f])$$

$$\text{Im}(Y[f]) = \text{Im}(X[f]) \times \text{Re}(H[f]) + \text{Re}(X[f]) \times \text{Im}(H[f]) \quad (4)$$

where Y is the output, X is the input, f is the frequency domain sample, H is the IR, Re is the real component, and Im is the imaginary component. According to this infrastructure, if data hiding is performed on the IR, and convolution is performed with the dry signal, the output will turn into a stego-audio.

3.2. Data hiding

During the data hiding phase, the method utilizes various manipulations such as signal attenuation, flipping, normalization, and FFT (Fast Fourier Transform). Fig. 3 clearly illustrates the data hiding phase of the proposed reverb hiding.

The proposed algorithm presents a detailed procedure for securely embedding and retrieving a secret message using reverb as a carrier. The procedure begins with loading the cover audio and impulse response (IR), which serve as the base signals for embedding. The secret message, which is often an image, is first encrypted, reshaped into a 1D array, and then its amplitude is reduced to ensure the embedded data remains imperceptible. The IR is flipped to allow the hidden message to begin from the end, optimizing space utilization. Metadata, including the dimensions of the secret message, is embedded into the initial 48 samples of the IR, followed by the appending of the reshaped message data into subsequent samples. The modified IR is normalized to match the loudness of the cover audio, thereby maintaining perceptual transparency. The normalized IR is then convolved with the cover audio in the frequency domain using the fast Fourier transform (FFT), resulting in a stego-audio signal. This signal combines the dry and reverberated signals in a specific mix ratio.

In the proposed method we use image as a secret message. The data hiding phase of the proposed method takes an original sound which is recorded in a studio environment. Then, the IR to be applied to that sound, and the message to be hidden (*i.e.*, message image) as input. The hidden image is encrypted using the XOR method with itself as the key. The size information of the secret message is then written to a binary array as a chunk with 1x48 size. Since the entire message is to be hidden at the beginning of the IR, a 10^6 reduction is performed to reduce the initial amplitude.

In the next step, gain normalization is applied to IR. Because, according to the technical report of the audio Engineering Society [31] the industry standard is to distribute album tracks at -14 LUFS (Loudness Units to full scale) or 0 dBFS (Decibels relative to full scale), but the loudness of the IR should be under control for this application. The

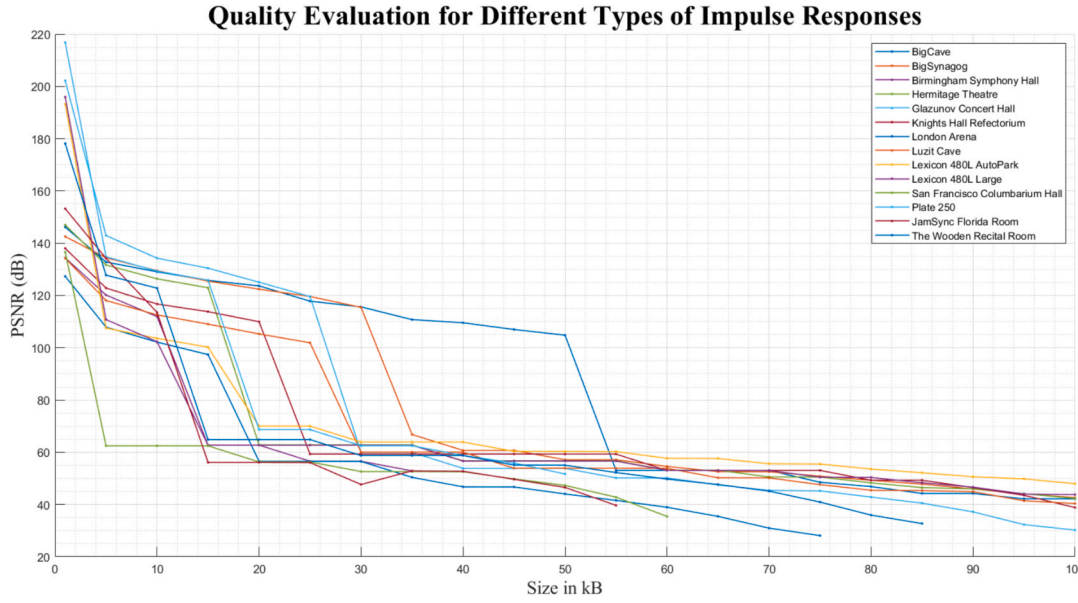


Fig. 6. Visualized results of the proposed method (32-bit depth floating-point) for different secret message sizes and IR's.

loudness of normalized IR must be calculated achieve this standard.

$$L_{IRN} = IR \times \frac{10^{(L_D - L_{IR})/20}}{20} \quad (5)$$

where L_{IRN} is the loudness of normalized IR, L_D is the loudness of dry signal, and L_{IR} is the loudness of the IR. The experiments yielded an IR of approximately -40 LUSF, which decreases the probability of stego-audio being clipped. Therefore, constants were also used to keep the L_{IRN} value around -40 LUSF.

During the data hiding phase, the message to be hidden is replaced by IR. IR is a sound that is not heard by the end user, i.e. the music listener, and often not even by the mix engineer. Therefore, the first 48 values of IR are utilized to indicate the message's length. Starting from the 49th value, the IR is replaced by the message as shown in Fig. 4. The capacity varies depending on the length of the IR and the sampling frequency.

In the final step of data hiding, the dry audio and IR are multiplied in frequency space, since processing in frequency space reduces computational cost. The stego audio of this stage is not in accordance with the principle of inaudibility, since the effect of the reverb is too high. So, a blend is made from 90 to 10 percent between the dry signal and the reverbed signal, like the mix parameter found in almost all reverb plug-ins. In this article, the 90/10 ratio is a preference inspired by its use in the music industry and reverb plug-ins. Even if adjusted to 80/20, the results would remain the same since our focus is on analyzing the reverb itself, not its distortion but its audibility. The use of reverb depends on the sound engineer's preference, and an acoustic guitar was chosen for its versatility across genres and broad frequency range.

The latest stego audio is obtained as given in equation (6):

$$\text{stegoaudio} = \frac{|\text{drysignal}| \times 90}{100} + \frac{|\text{reverbedsignal}| \times 10}{100} \quad (6)$$

The reverbed signal is also the part that contains the hidden message. But this signal cannot be used because it is 100 % wet. So, it has to be blended with a dry signal, as if it were a parallel track, so that it sounds suspiciously dry. Fig. 3 shows the flowchart for the entire data hiding process.

3.3. Data extracting

The data extraction process is simpler than data hiding. The message

retrieval procedure begins with the extraction of the embedded message from the stego-audio via deconvolution. The original secret message is then reconstructed by decoding and decrypting the extracted data. This enhanced step of reducing the image amplitude ensures that the embedding process minimizes the perceptual impact on the audio signal while preserving security and quality.

It is assumed that the receiving party has access to both the dry signal and the coefficient used for normalization. The first step in this stage is to obtain a 100 % wet (reverbed) signal from the blended (stego) signal, which can be achieved using the following formula:

$$\text{reverbedsignal} = \left(\text{stegoaudio} - \frac{|\text{drysignal}| \times 90}{100} \right) \times 10 \quad (7)$$

The wet signal is obtained by applying IR to the dry signal and must be obtained through deconvolution since the part containing the message is in the IR. IR can be obtained by operating in the frequency space as follows:

$$\text{Re}(H[f]) = \frac{\text{Re}(Y[f])}{\text{Re}(X[f])} \quad (8)$$

where H is the IR, Y is the output, X is the input, f is the frequency domain sample, and Re is the real component. Finally, for $\text{Re}(H[f])$, the inverse of the change resulting from the normalization process is applied. The length of the message is determined from the first 48 values of the resulting IR. From the 49th value onwards, the message is extracted, and decryption is performed. Fig. 5 shows the flowchart for the entire data extracting process.

4. Experimental results

The experimental results obtained with several IRs from a variety of sources, including Small Rooms, Medium Rooms, Theaters, Arenas, Concert Halls, Medium Halls, and Churches. The experimental set also contained a plate device and IRs from the legendary Lexicon 480L. All these IRs are used for obtaining of the reverbed audios. The Reverb Hiding – Impulse Responses dataset generated for this paper is accessible in the Kaggle platform [35].

The Waves IR Library and the sweep file are used for the experimental results which are freely accessible in [37]. During this capture, the exponential (log) sweep sine method is used to eliminate nonlinear components and minimize the total harmonic distortion [38,39]. The

Table 2
PSNR (dB) values of 32-bit depth fixed-point Stego Audio for various IRs with different sizes of secret message.

Secret Message Size	PSNR (dB)									
	Big Cave	Big Synagog	Birmingham Symphony Hall	Hermitage Theatre	Glazunov Concert Hall	Knights Hall Refectorium	London Arena	Luzit Cave	Lexicon 480L AutoPark	Lexicon 480L Large
1 kB	87.65	95.08	95.08	90.62	85.26	95.08	89.38	78.45	95.08	79.57
5 kB	87.64	95.08	95.08	90.59	85.26	95.08	89.38	78.45	95.08	79.57
10 kB	87.64	95.08	95.08	90.52	85.26	95.08	89.38	78.45	95.08	79.57
15 kB	87.64	95.08	95.08	90.41	85.26	95.08	89.38	78.45	95.08	79.57
20 kB	87.63	94.99	66.06	90.15	85.23	87.61	76.09	76.89	69.71	74.07
25 kB	87.58	94.72	65.04	89.50	85.06	81.07	67.28	74.56	67.21	70.10
30 kB	87.54	93.97	58.54	64.07	62.59	76.15	57.01	56.86	61.53	62.65
35 kB	87.33	68.50	54.70	64.06	62.58	61.49	52.97	50.90	58.62	60.90
40 kB	87.24	68.47	48.00	64.02	62.55	61.07	48.37	50.75	55.49	59.38
45 kB	86.93	68.45	41.28	58.02	56.53	60.28	43.90	48.67	53.81	57.23
50 kB	86.52	68.29	36.42	54.46	56.35	54.83	38.80	48.24	51.11	54.41
55 kB	63.78	62.33	33.39	51.91	52.33	54.25	32.37	45.02	47.10	51.63
60 kB	63.76	61.39	–	49.88	46.80	53.70	28.92	43.70	44.03	48.49
65 kB	63.74	59.77	–	49.76	44.62	53.48	27.95	42.66	42.36	46.86
70 kB	57.74	54.37	–	49.24	38.57	50.68	26.19	40.29	39.33	44.26
75 kB	57.71	51.40	–	46.69	34.45	50.52	25.26	37.65	37.44	41.44
80 kB	57.63	45.30	–	40.24	29.25	47.97	–	33.89	33.96	38.54
85 kB	57.47	41.19	–	37.01	27.50	46.10	–	28.74	32.94	35.79
90 kB	53.91	36.57	–	33.49	26.09	37.15	–	26.35	31.60	32.95
95 kB	53.50	33.23	–	30.45	25.26	32.03	–	26.35	30.62	30.21
100 kB	52.83	31.73	–	28.38	24.99	30.04	–	24.62	29.78	28.90

Table 3
PSNR (dB) values of 24-bit depth fixed-point Stego Audio for various IRs with different size of secret message.

Secret Message Size	PSNR (dB)									
	Big Cave	Big Synagog	Birmingham Symphony Hall	Hermitage Theatre	Glazunov Concert Hall	Knights Hall Refectorium	London Arena	Luzit Cave	Lexicon 480L AutoPark	Lexicon 480L Large
1 kB	89.35	95.08	95.06	89.91	85.26	94.98	88.66	78.45	95.08	79.57
5 kB	89.35	95.08	94.47	89.89	85.26	92.68	85.62	78.26	95.08	79.57
10 kB	89.35	95.08	66.21	89.83	85.25	89.04	62.5	77.81	95.08	79.57
15 kB	89.35	95.08	66.19	89.73	85.24	86.68	62.43	77.11	95.08	79.57
20 kB	89.35	94.99	60.13	89.51	85.19	83.22	56.43	75.76	69.71	74.07
25 kB	89.35	94.72	59.82	88.95	85.02	79.63	56.13	73.87	67.21	70.1
30 kB	89.35	93.97	55.82	64.07	62.59	75.6	51.79	56.85	61.53	62.65
35 kB	89.34	68.5	53.31	64.06	62.58	61.47	50.16	50.9	58.62	60.9
40 kB	89.34	68.47	47.38	64.02	62.55	61.05	46.65	50.74	55.49	59.38
45 kB	89.33	68.45	40.95	58.02	56.53	60.26	42.93	48.67	53.81	57.23
50 kB	64.02	68.29	36.21	54.46	56.35	54.83	38.19	48.24	51.11	54.41
55 kB	64.02	62.33	33.22	51.91	52.33	54.24	32.01	45.02	47.1	51.63
60 kB	62.02	61.39	–	49.88	46.8	53.7	28.64	43.7	44.03	48.49
65 kB	57.93	59.77	–	49.76	44.62	53.47	27.69	42.66	42.36	46.86
70 kB	57.93	54.37	–	49.24	38.57	50.68	25.96	40.29	39.33	44.26
75 kB	57.93	51.4	–	46.69	34.45	50.52	25.04	37.65	37.44	41.44
80 kB	54.33	45.3	–	40.24	29.25	47.97	–	33.89	33.96	38.54
85 kB	54.33	41.19	–	37.01	27.5	46.1	–	31.44	32.94	35.79
90 kB	54.33	36.57	–	33.49	26.09	37.15	–	28.74	31.6	32.95
95 kB	51.75	33.23	–	30.45	25.26	32.03	–	26.35	30.62	30.21
100 kB	49.74	31.73	–	28.38	24.99	30.04	–	24.62	29.78	28.9

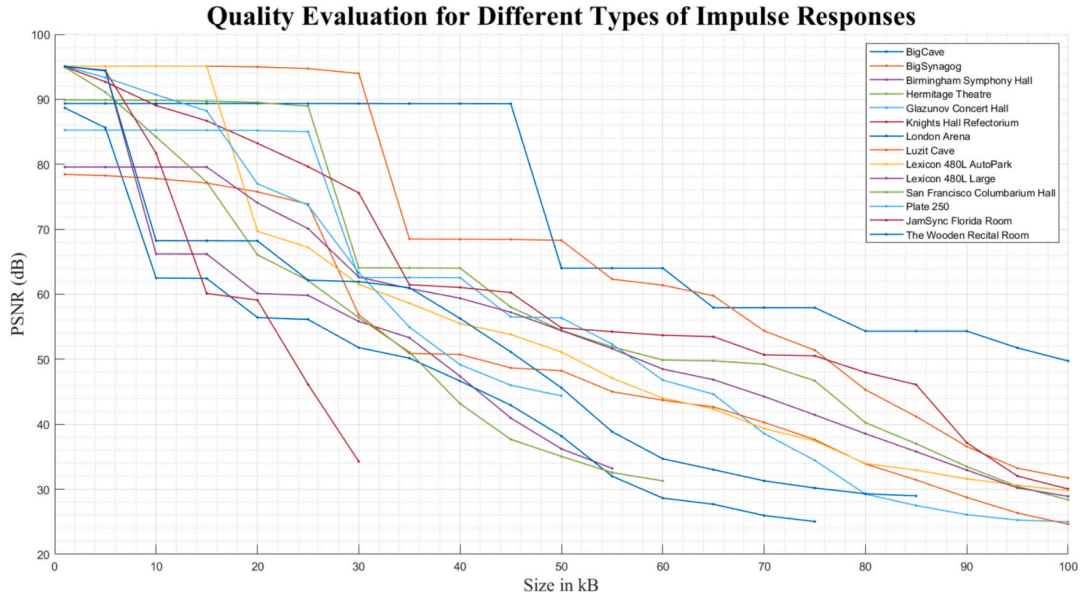


Fig. 7. Visualized results of the proposed method (24-bit depth fixed-point) for different secret message sizes and IR's.

exponential sweep sine method is used for linearity. During this capture, the exponential sweep sine method is used to eliminate nonlinear components and minimize the total harmonic distortion. In the method of [39] log (exponential) sine sweep is used to obtain the impulse response from the recorded audio. The properties of the selected sine sweep signal are also given in the relevant article. The inverse filter of the sine sweep used in deconvolution is used. This inverse filter is used to suppress non-linear harmonics. Then, the non-linear behavior of the IR obtained with linear deconvolution is suppressed. So, all IRs are linear since it is used the Waves IR Library capture method. Therefore, in our proposed study, it is worked with signals that are already linearized. Thus, the technical processes of our study follow the basic principles of convolution.

The results were evaluated with several audio quality approaches. The numerical results compared in terms of the peak signal-to-noise ratios (PSNR). The PSNR is one of the objective quality evaluation metrics of the audio-based data hiding methods. It is important to note that both the units of PSNR and dynamic range units are in dB. But they are different terms for signals. PSNR defines the quality of the modified signal, where the dynamic range for audio refers to the difference between the loudest and quietest sounds a system can accurately represent or reproduce. It is typically measured in decibels (dB). The calculation of the peak signal-to-ratio requires determination of the mean squared error between original and distorted signal, which is derived using the following formula:

$$MSE = \frac{1}{N} \sum_{i=1}^N (S_i - S'_i)^2 \quad (9)$$

where N is the number of samples, S and S' are the original cover audio and stego audio signal values of related samples respectively.

Then the following formula is used to calculate PSNR:

$$PSNR = 10 \times \log_{10} \left(\frac{(2^b - 1)^2}{MSE} \right) \quad (10)$$

where b is bit depth of the audio. According to the above formulas, the PSNR of same signals will be infinite. However, it should be noted that, since the unit of both PSNR and dynamic range is dB. But here dB is meant in quality assessment is the result of PSNR.

An effective data hiding should have a high PSNR for stego objects without any loss or compression [3]. If lower PSNR results are obtained,

the human ear detects distortions in the stego audio. While PSNR is important evaluation metrics for data hiding, also visual results compared in terms of oscillograms, frequency spectrum analysis, spectrograms, cepstrogram analysis, histogram analysis and autocorrelation analysis. Each of these analysis methods helps with a unique purpose in understanding and characterizing audio signals, especially in applications such as audio processing, steganography, or forensic analysis. The last comparison is given for the statistical features of the cover and stego audios. To avoid the muddy mixing of low and high frequencies due to nonlinear reverbs, a band-pass filter is generally used in before reverb [36]. However, this process was not applied in the experiments in order to examine the full range of variation in the reverb.

Experimental results obtained with MATLAB® R2023b by using the computer with these technical specifications; Intel® Alder Lake Core™ i5-12500H 12C/16 T; 18 MB L3; E-CORE Max 4.5GHZ, nVIDIA® GeForce® RTX3050 TI Max Performance 4 GB GDDR6 128-Bit, 32 GB DDR4 1.2 V 3200 MHz, 500 GB PCIe M.2 2280 3.0 x4.

In the first experiment, the PSNR values of different impulse responses (IRs) were compared, depending on their capacity. To calculate PSNR, the dry signal processed with an IR (without any data embedding) was compared to the same signal processed with different amounts of data embedded in the IR. In other words, the reverberated cover audio was compared to the reverberated stego audio. The secret message was embedded within the IRs used for reverberation. The IRs had different lengths, and in some cases, longer messages were intentionally embedded to maintain an acceptable PSNR level, even when using the maximum possible hiding capacity.

Even when the PSNR values were relatively low, these reductions did not cause the listeners to notice any issues which are expected. This is significant because, despite the low PSNR values in mathematical terms, the resulting audio, as described by the framework presented in this study, was still perceived as high quality by listeners. The PSNR comparison results of the proposed reverb-based data hiding method are shown in Table 1, for secret message sizes ranging from 1 kilobyte to 100 kilobytes. The maximum length of the secret message depends directly on the length of the IR, not on the dry signal. According to the results in Table 1, the PSNR values obtained are quite high. Therefore, the distortions in the videos are too low to be perceived by the human auditory system.

The results of Table 1 visualized in Fig. 6 to see the PSNR results for the tests using the 14 different IRs. Fig. 6 shows that there is resistance to

Table 4
RT, DRR, C50, C80, and EDT for various IRs before data hiding.

Before Data Hiding														
	Big Cave	Big Synagog	Birmingham Symphony Hall	Hermitage Theatre	Glazunov Concert Hall	Knights Hall Refectorium	London Arena	Luzit Cave	Lexicon 480L AutoPark	Lexicon 480L Large	San Francisco Columbarium Hall	Plate 250	JamSync Florida Room	The Wooden Rectal Room
RT	8.72	8.92	5.31	7.53	6.11	12.00	7.73	12.63	15.59	12.39	5.84	5.13	3.15	6.63
DRR	-30.84	-31.45	-27.85	-30.41	-29.60	-27.37	-29.46	-31.99	-31.19	-31.75	-29.86	-29.59	-25.51	-30.56
C50	-18.04	18.51	-14.73	-17.73	-16.59	-14.24	-16.76	-18.48	-20.47	-21.67	-16.55	-16.52	-11.92	-17.67
C80	-16.06	-16.39	-12.62	-15.50	-14.50	-12.11	-14.77	-16.40	-18.51	-19.07	-14.44	-14.42	-9.71	-15.59
EDT	13.24	26.93	12.18	17.69	14.11	8.23	16.89	11.53	42.36	26.61	16.85	15.62	5.80	21.26

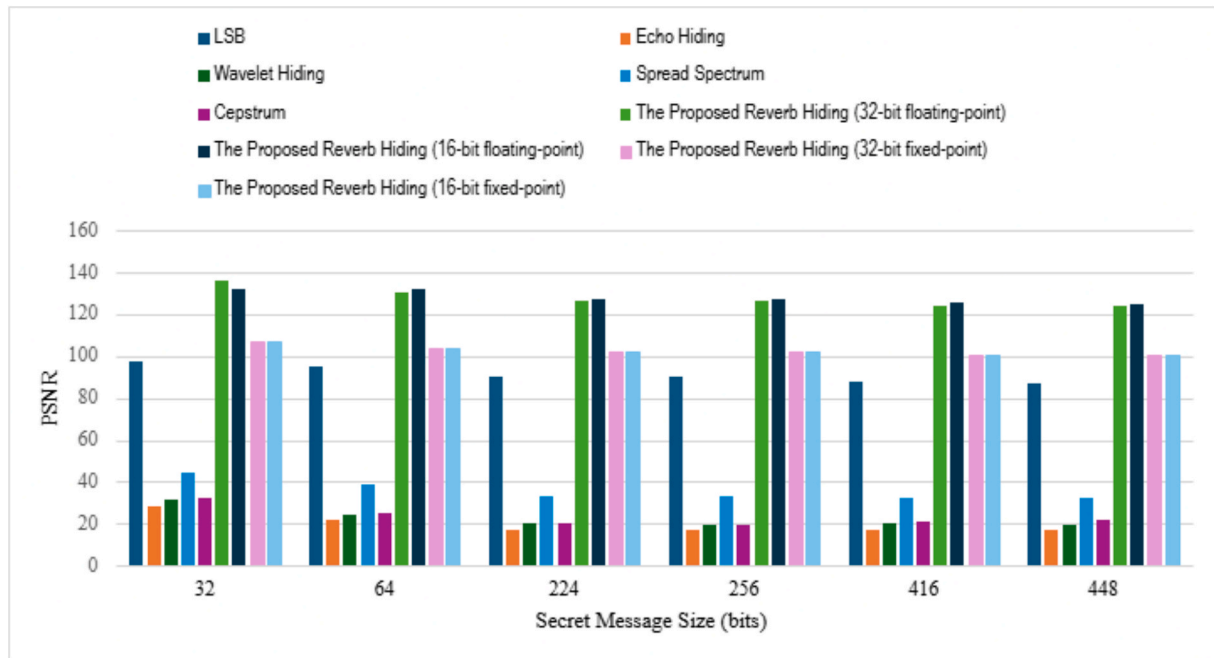
Table 5
RT, DRR, C50, C80, and EDT for various IRs after 50 kb size of data hiding.

After Data Hiding														
	Big Cave	Big Synagog	Birmingham Symphony Hall	Hermitage Theatre	Glazunov Concert Hall	Knights Hall Refectorium	London Arena	Luzit Cave	Lexicon 480L AutoPark	Lexicon 480L Large	San Francisco Columbarium Hall	Plate 250	JamSync Florida Room	The Wooden Rectal Room
RT	8.72	8.92	5.14	7.52	6.11	11.88	7.93	12.53	16.20	11.90	6.40	5.61	2.36	6.66
DRR	-30.84	-31.45	-27.85	-30.41	-29.60	-27.37	-29.46	-31.99	-31.19	-31.75	-29.86	-30.56	-25.51	-30.56
C50	-18.04	-18.51	-14.73	-17.73	-16.59	-14.24	-16.76	-18.48	-20.47	-21.67	-16.55	-16.52	-11.92	-17.67
C80	-16.06	-16.39	-12.62	-15.50	-14.50	-12.11	-14.77	-16.40	-18.51	-19.07	-14.44	-14.42	-9.71	-15.59
EDT	13.24	26.93	12.18	17.69	14.11	8.27	16.89	11.53	42.36	26.61	16.86	15.63	5.79	21.26

Table 6

The PSNR (dB) comparison with the state-of-the-art audio steganography methods.

Secret Message Size (bits)	LSB	Echo	Wavelet	S. Spectrum	Cepstrum	The Proposed Reverb Hiding (32-bit floating-point)	The Proposed Reverb Hiding (16-bit floating-point)	The Proposed Reverb Hiding (32-bit fixed-point)	The Proposed Reverb Hiding (16-bit fixed-point)
32	97.14	27.98	31.82	44.09	32.20	136.02	131.98	106.87	106.88
64	95.28	22.20	24.23	38.38	24.99	129.99	131.98	104.21	104.23
224	90.28	16.67	19.97	32.70	20.19	126.02	127.21	101.97	101.98
256	90.14	16.69	19.78	32.80	19.50	126.01	127.21	101.87	101.88
416	88.13	16.61	20.03	32.53	21.38	123.97	125.54	100.32	100.32
448	87.26	16.66	19.56	32.40	21.94	123.71	124.99	100.25	100.26

**Fig. 8.** The PSNR comparison with the state-of-the-art audio steganography methods.

the sound quality at 120 dB and 60 dB PSNRs. However, some IRs resisted longer at much larger amounts of data, apart from the general appearance. Some IRs sizes are short, and their capacities are fully used with the secret messages. Some other IRs are longer enough to embed secret data until 100 kB. It is necessary to test a large data set to select the appropriate IR if someone needs specific sizes. The present study investigates the differences between fixed-point and floating-point audio files. To this purpose, data hiding was performed with a 32-bit depth fixed-point audio file in Table 2.

The same experiment was performed with 24-bit depth fixed-point audio files, which are commonly used in music studios [40]. In this way, the applicability to the music industry was also tested. The results are presented in Table 3 and Fig. 7. Similar to the 32-bit audio files, the result of the 24-bit audio files are also present a good quality for output of the data hiding process.

The fundamental nature of steganography is such that it causes distortion to the carrier used. We have shown our results with popular metrics in the field of audio steganography. Also, we tested the reliability of our results by including important audio reverb measure metrics, the calculation details of these metrics can be found in [33]. In order to explore whether this distortion alters the characteristics of IR, a further experiment was conducted. A set of metrics was selected for the purpose of examining the characteristics of the reverb. Reverb time (RT) denotes the duration of the reverb. C50 and C80 refer to the correlation of energies in the first 50 ms (for speech) and first 80 ms (for music), respectively. Direct-to-reverberant ratio (DRR) refers to the ratio between direct sound energy and the reflected sound energy from the

environment. early decay time (EDT) is associated with perceived running reverberation and directly correlates with the damping of the reverb tail [33]. The experiment was performed for the original IR and for the IR with 50 kb of data hidden. The results of the experiment are given in Table 4 and Table 5.

When the results of Table 4 and Table 5 are compared, the impact of data hiding on the RT, DRR, C50, and C80 values of IR characteristics was not observed to be unfavorable. In the data hiding phase, the process of data hiding begins from the end of the IR, maintaining the initial characteristics of the beginning as much as possible. This approach is intended to preserve the energy levels in the first 50 ms and 80 ms as similar as possible. The alteration caused by data hiding was found to be limited to the RT value and was not significant enough to impact the character in the tail of the reverb. It is important to note that sound engineers generally prefer different reverb time for different recordings.

The modification in the tail, a critical element of the reverb, was examined through the utilization of EDT. At this point in the analysis, the alteration in EDT remains below the ISO 3382 standard [34] of 5 % for just noticeable difference (JND).

This paper proposed a novel data hiding method for audio signals with convolution reverb by using artificially generated IRs. The proposed method achieves good imperceptibility in terms of PSNR. The advantage of the proposed method is proven by comparison with known state-of-the-art audio steganography methods LSB, Echo Hiding, Wavelet, Spread Spectrum and Cepstrum [30]. The proposed reverb hiding method compared for secret messages with different sizes. The cover audio of the proposed method was sampled at both 32-bit depth

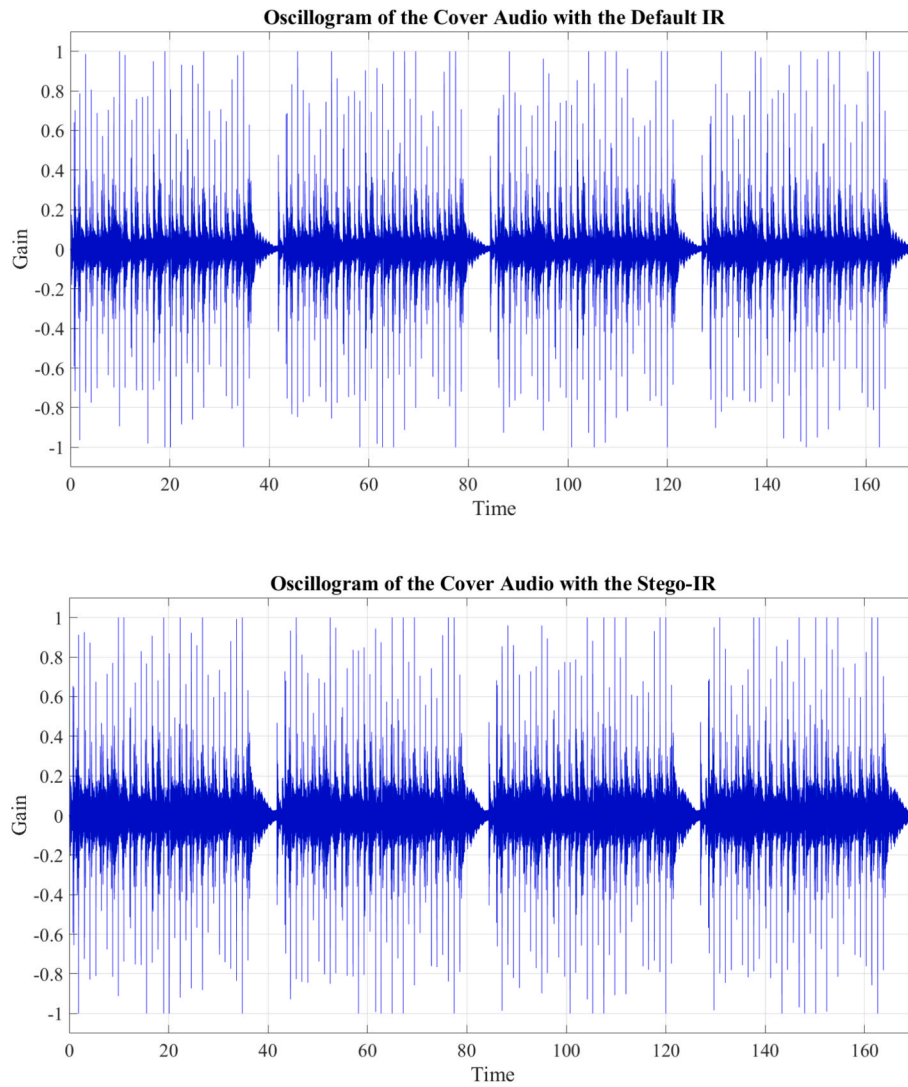


Fig. 9. Oscillograms of the cover and stego audios.

and 16-bit depth in this experiment to measure the effect of bit depth. Because, the 16-bit format offers a balanced solution between energy efficiency and high sound quality, while the 32-bit format provides the theoretically widest dynamic range, making it the preferred choice for high-end applications [32]. The 16-bit depth audio file is also floating-point in order to provide diversity of audio formats in the experimental results. The comparison results are shown in Table 6, and it is crucial to note that the values presented in the table correspond to peak signal-to-noise ratio (PSNR) metrics, not to dynamic range. It has been observed that as the amount of data hidden in 16-bit and 32-bit audio files increases, they reach the same PSNR values. This phenomenon due to the fact that at very small distortions, the quantization error becomes larger than the distortion itself. However, as the amount of hidden data increases, the distortion caused by data hiding becomes more dominant than the quantization error. Consequently, whether fixed-point or floating-point is employed, the proposed method does not generate a significant difference in bit space, as it interferes with the impulse response used during reverb rather than the audio perceived by the end user. The results of Table 6 are also visualized in Fig. 8. The results strongly indicate that, regardless of resolution, the reverb hiding method has a much higher PSNR than its competitors when hiding relatively secret messages.

In the first step of the experimental results imperceptibility of the proposed method evaluated with several capacity conditions. The

experiments show the proposed reverb hiding method has remarkable results when compared with the most popular literature methods of audio steganography.

In the second parts of the experiments, the visual results compared in terms of oscillograms, frequency spectrum analysis, spectrograms, cepstrogram analysis, histogram analysis and autocorrelation analysis. The comparison focused on the effect of the data hiding with reverb method. For a fair comparison, all the cover audio denotes dry audio signal with default IR signal and the stego audio denotes the audio signal with data embedded IR. All comparison for visual analysis done with two different audio files. The first audio is the convolution of the original St. Paul Chapel's IR with the cover audio. The second audio is the convolution of the stego St. Paul Chapel's IR with the cover audio where 100 kb secret message was hidden in this IR. The secret message size is about 17 % of the IR. The cover audio contains only an acoustic guitar recorded with a single microphone, without any processing such as compression, limiting, or clipping. The four repetitions simulate a real song structure.

Fig. 9 shows the oscillogram comparison of the proposed method. An oscillogram is a graphic representation of audio data that shows how amplitude and time relate to one another. An oscillogram's horizontal axis displays seconds of time, and its vertical axis represents amplitude. The tool lets you view and evaluate how amplitude changes over time from loud to soft. Each point along the waveform reflects relative amplitude in decibels. Although oscillograms alone might not be enough

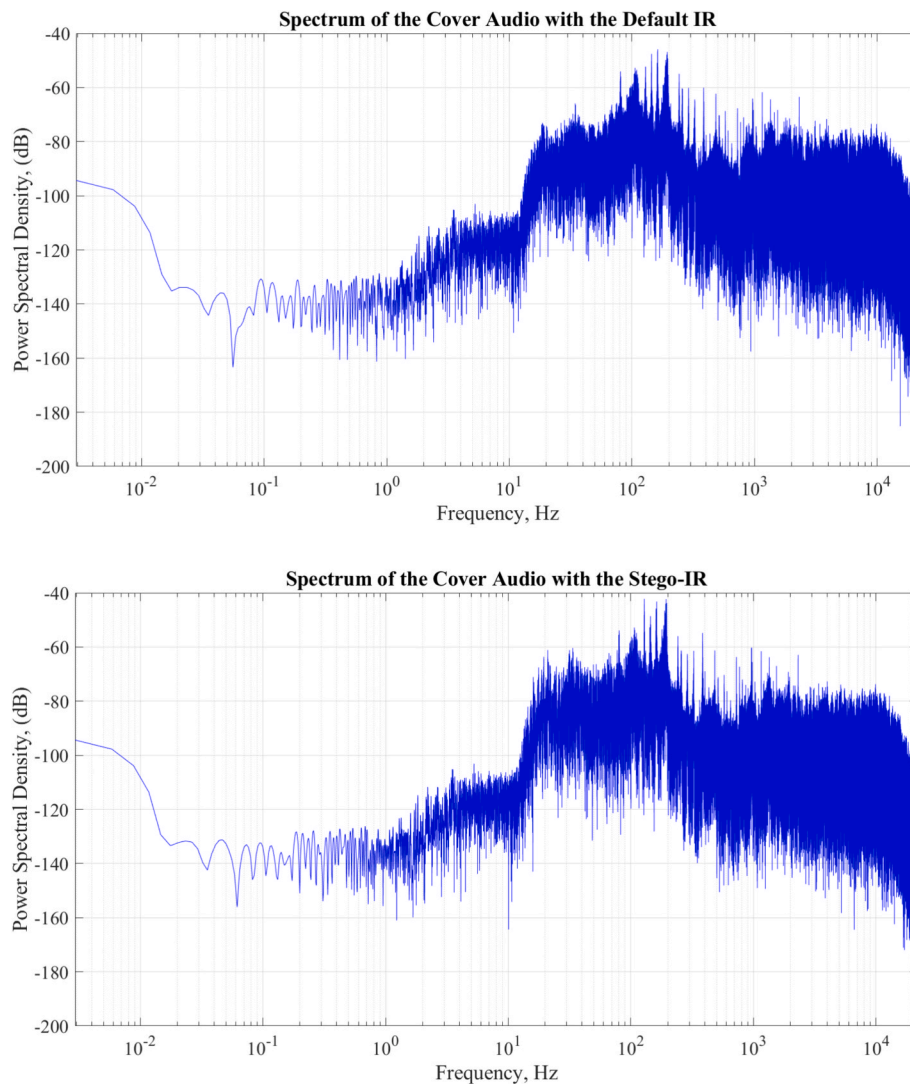


Fig. 10. Frequency spectrum analysis of the cover and stego audios.

for perceptual evaluation, they remain a valuable tool when combined with more perceptually-oriented analyses. Fig. 8 shows that there were no visual differences found in the oscillograms of the two audio files.

The second visual analysis tested for the frequency spectrum and the results given in Fig. 10. As expected, due to the presence of only one acoustic guitar in the cover audio, there was concentration in the high-mids in Fig. 10. Low frequencies have intentionally been left blank to clearly demonstrate any potential distortion caused by data hiding. Fig. 10 indicates that there are changes in the frequency spectra below 1 Hz, but no noticeable changes above 4 Hz, the lower threshold of most headphones and speakers.

The third comparison tested for spectrogram and the results given in Fig. 11. A spectrogram shows frequency and magnitude changes over time. It displays elapsed time in a left-to-right format similar to an oscillogram, but it also includes a frequency dimension. Amplitude is shown by color variations that indicate the intensity of the vocalization that occurred, while frequency is shown on the vertical axis in hertz. According to Fig. 11, the time-dependent frequency changes remained constant in the spectrogram analysis as expected. However, the characteristic slap strumming, which is preferred in percussive playing, is clearly visible through the vertical lines. Notably, these lines remain unchanged in the spectrogram analysis.

Fig. 12 demonstrates the results of cepstrogram analysis, which is often used to analyze time-dependent sound effects. The frequency

distributions show that the hidden data does not change the character of the cover audio or the reverb effect itself.

The histogram analysis in Fig. 13 illustrates that the distribution of gain values remains essentially unchanged, with the shape of the histogram closely matching a normal distribution. This finding indicates that the data hiding process did not cause a substantial impact on the statistical properties of the signal.

Finally, an autocorrelation analysis is performed in Fig. 14. This analysis is particularly important against the possibility of periodic pattern formation due to hiding. The autocorrelation data shows that the use of data-hidden IR did not produce a periodic pattern or significant time-dependent noise.

When Fig. 8 to Fig. 14 are evaluated, it is seen that there is no visual difference between the graphs obtained before and after data hiding. The original and stego audio signals produce similar graphics across analyses for oscillograms, frequency spectrum analysis, spectrograms, cepstrogram analysis, histogram analysis and autocorrelation analysis. It indicates that the modifications have introduced minimal distortion to the audio. It means that the perceptual and acoustic properties of the original signal have been preserved, ensuring perceptual transparency and maintaining the signal's integrity. The preservation of critical features, such as amplitude, frequency content, and temporal structure, implies that the modifications are subtle and unlikely to be detected by human hearing. Furthermore, consistent histogram and autocorrelation

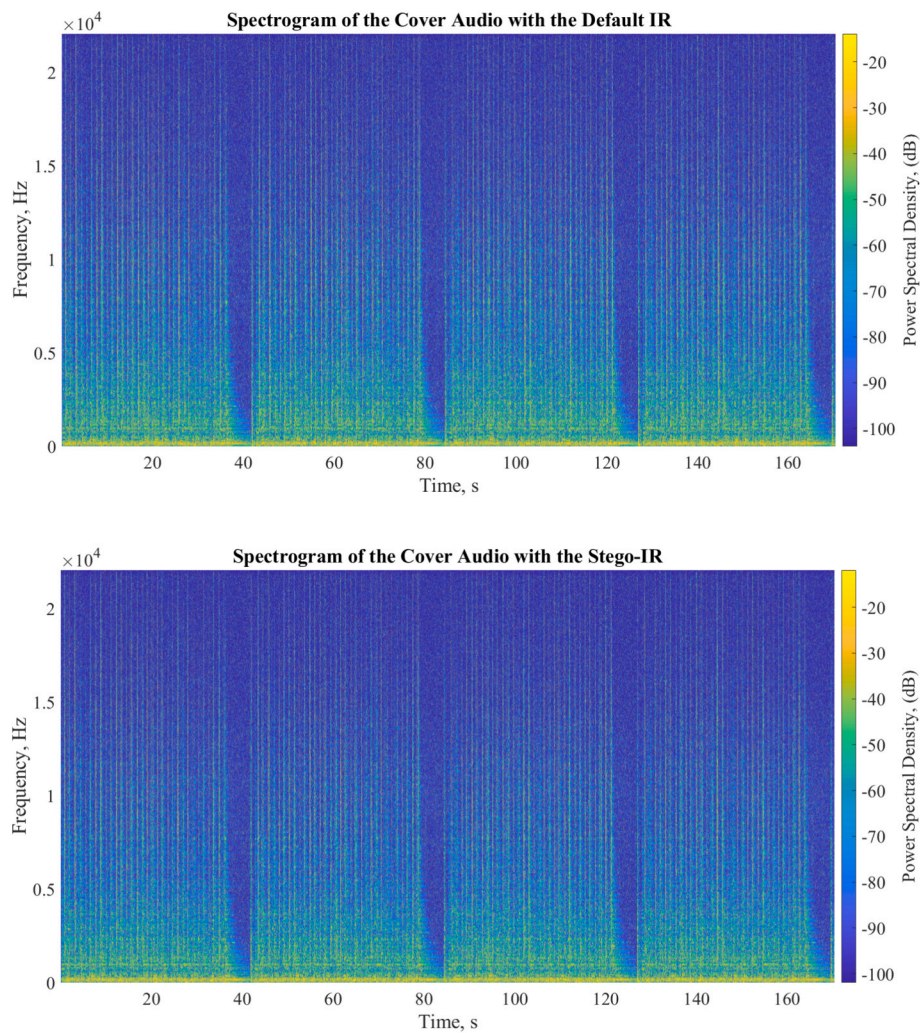


Fig. 11. Spectrograms of the of the cover and stego audios.

results show that the signal's statistical features, such as dynamic range and periodicity, remain unchanged. Overall, this demonstrates the modification method's robustness, guaranteeing that the desired modifications are implemented without harming the quality or features of the original audio.

The last comparison of the experimental results is given for the statistical features of the cover and stego audios. In addition to the visual analysis, the minimum and maximum values, mean value, RMS value, dynamic range, crest factor value and autocorrelation time of the two audio signals are shown in Table 8. According to the results of Table 8, there is no significant difference between cover and stego files in terms of the statistics except of the dynamic range.

Dynamic range is the difference between the highest and lowest points of a sound's amplitude. CD quality is generally considered to be 96 dB, although many modern audio interfaces can play audio above 110 dB. In Stego-audio, the amplitude is reduced due to the normalization process on the IR. By normalizing with a higher LUFS target, the dynamic range can be increased to the same level as the cover audio.

5. Conclusion

This study proposes a new framework for audio steganography, which plays a significant role in applied music technologies. Applied music technologies have opened a new field through the use of steganography methods. Other time-based effects can also benefit from the use of similar applications, not just reverb. The proposed method

embeds secret message into the impulse response of the reverb signal. The impulse response convoluted with the real audio and the stego audio obtained. We tested our novel reverb hiding method with several reverb IR of natural environment. Due to the size of the IRs are varied different sizes of the secret messages embedded into them. The proposed method aims to reduce the losses or changes after embedding the secret message into the audios. PSNR is one of the important measures of quality degradation. All studies in the literature on the change before and after data hiding use PSNR and similar objective metrics. The bit depth has significant importance in the audio files used for data embedding. Thanks to the convolution process, PSNR values are high and the proposed reverb hiding achieves superior quality in terms of PSNR of the stego audios at varying bit depths. Also, proposed method has significant superior when compared with state-of-the-art audio steganography methods LSB, Echo, Wavelet, Spread Spectrum and Cepstrum. Also, the proposed method tested the reliability of the results by including important audio reverb metrics RT, DRR, C50, C80, EDT, and JND. The proposed method achieves a robust performance in terms of these metrics. Various audio analysis graphs are used to show that there is no significant difference in stego-audio because of the proposed method. It is acceptable to have minor changes in these graphs. However, no significant change is observed in the main graphs obtained with the results of the proposed method. The effect of the reverb hiding tested by comparing audio quality control techniques such as scalogram, spectrograms, autocorrelation analysis, etc. The reverb method is very popular for the music industry, so in the future works reverb hiding will

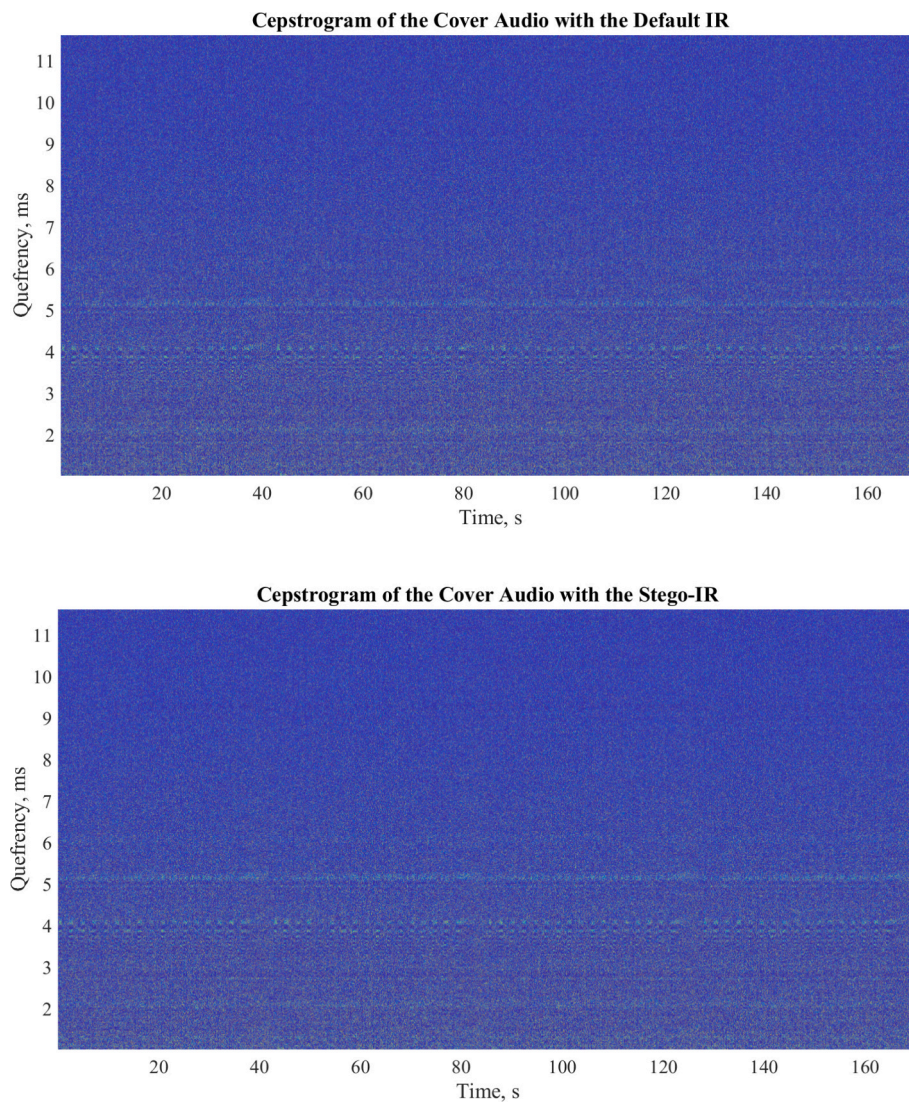


Fig. 12. Cepstrogram analysis of the cover and stego audios.

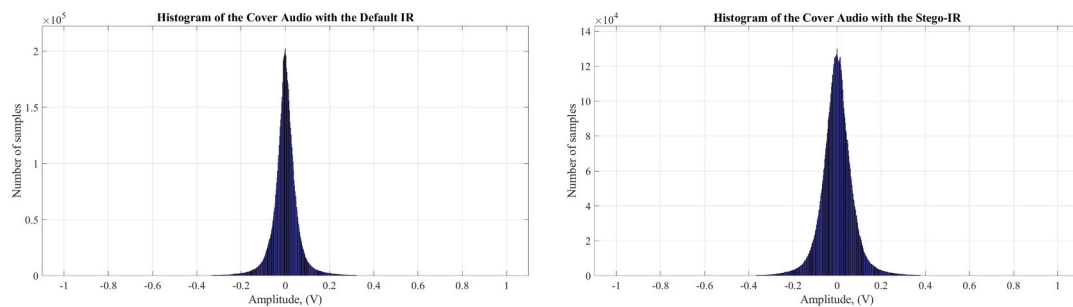


Fig. 13. Histogram analysis of the cover and stego audios.

become more popular for steganography science. So, our future work will include various frequency-based algorithms for reverb hiding.

CRedit authorship contribution statement

Ali Erdem Altınbaş: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mehmet Zeki Konyar:** Writing – review & editing,

Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

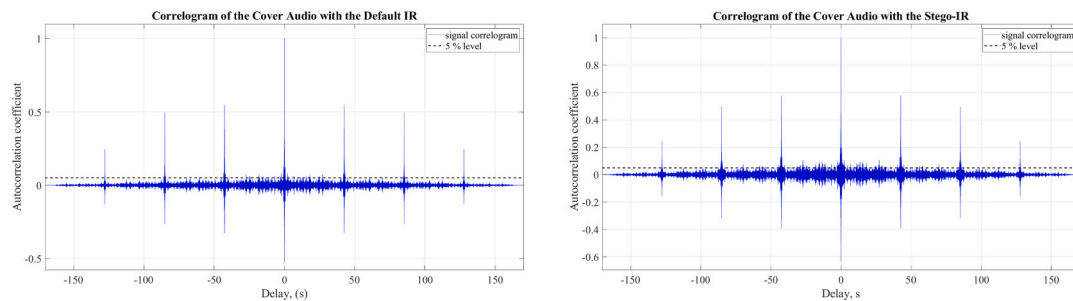


Fig. 14. Autocorrelation analysis of the cover and stego audios.

Table 8

Statistics of the cover and stego audios.

	Cover Audio	Stego Audio
Maximum Value	0.99998	0.99997
Minimum Value	-0.99998	-1
Mean	-1.4207×10^{-19}	-1.5298×10^{-19}
RMS	0.062638	0.062638
Dynamic Range	96.3849 dB	90.309 dB
Crest Factor	24.0631 dB	24.0633 dB
Autocorrelation Time	127.9369 s	127.9369 s

Data availability

Data will be made available on request.

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